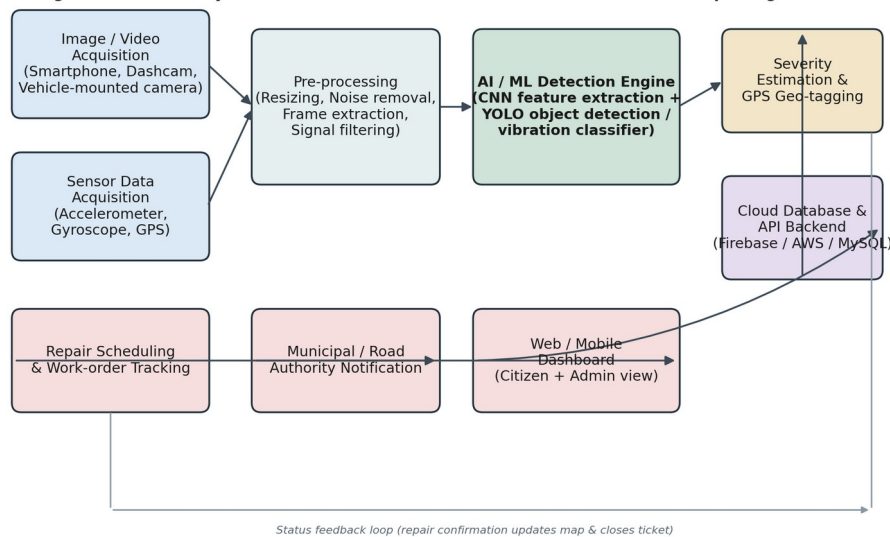


A Technical Review Paper on

StreetGuard AI Pothole Reporter

A Review of Existing AI / ML-Based Pothole Detection and Reporting Systems

Fig. 1 Generalised System Architecture of an AI-Based Pothole Detection & Reporting Platform



Domain: Artificial Intelligence / Machine Learning
Project Documentation / Review Paper

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Abstract

Potholes are among the most common and hazardous forms of pavement distress, causing vehicle damage, traffic accidents, and heavy financial losses to both citizens and civic authorities. In much of the world, pothole identification is still a manual and reactive process: road inspectors survey stretches of highway on foot or by vehicle, or repairs are triggered only after a citizen files a complaint. This approach is slow, inconsistent, and difficult to scale across large road networks. Over the past decade, a class of Artificial Intelligence (AI) and Machine Learning (ML) based systems has emerged to automate this process. Cameras mounted on vehicles or smartphones, combined with Convolutional Neural Networks (CNNs) and single-shot object detectors such as YOLO (You Only Look Once), can identify potholes in real time from ordinary video; complementary approaches use accelerometer and gyroscope sensors to sense road-surface vibration; and crowdsourced citizen-reporting platforms combine GPS geo-tagging with cloud dashboards to route verified reports to the correct municipal authority.

This paper is a technical review, prepared as project documentation for "StreetGuard AI Pothole Reporter", that studies several existing AI/ML-based pothole detection and reporting systems documented in academic literature and deployed applications, rather than proposing a new implementation. It examines the underlying theory (CNN feature extraction, YOLO-based object detection, vibration-signal thresholding, and detection-evaluation metrics such as Intersection over Union and mean Average Precision), the system architecture and methodology common to these platforms, their step-by-step working process, representative datasets such as the multi-national RDD2022 road-damage dataset, and performance figures reported in published work. The review closes with a discussion of the advantages, benefits, real-world applications, current limitations, and future scope of AI-based pothole detection, providing a theoretical foundation for the design of the StreetGuard system.

Keywords: Pothole Detection, Deep Learning, Convolutional Neural Network, YOLO, Object Detection, Smart City, Road Safety, Crowdsourcing, Computer Vision, IoT.

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1. Introduction

Roads form the backbone of a nation's transport and economic infrastructure, and the condition of the road surface directly affects safety, travel time, vehicle life, and logistics costs. A pothole is a bowl-shaped depression that forms when water seeps into cracks in the pavement, weakens the sub-base, and repeated traffic loading then breaks up the weakened surface layer. Poor material quality, inadequate drainage, seasonal freeze-thaw or monsoon cycles, and delayed maintenance all accelerate this process. Left unattended, a small crack can widen into a hazardous cavity within weeks, particularly during the monsoon season in tropical countries such as India.

Traditionally, pothole identification has depended on manual visual inspection by road engineers walking or driving along a stretch of road, or on reactive systems in which repairs are triggered only after a citizen telephones a municipal office or lodges a written complaint. Both approaches are slow, labour-intensive, and difficult to scale to the size of a modern road network; they also tend to prioritise repairs by complaint frequency rather than by actual severity, so that dangerous but under-reported potholes can remain unattended for long periods [15].

Over roughly the last fifteen years, this manual model has begun to be supplemented and, in some deployments, replaced by AI/ML-based systems. Early efforts such as the City of Boston's "Street Bump" used a smartphone's built-in accelerometer and GPS to sense the jolt of a wheel striking a pothole and to crowdsource the resulting readings from many vehicles [17]. More recent systems use computer vision and deep learning: a camera mounted on a vehicle dashboard or held by a pedestrian captures video of the road surface, and a Convolutional Neural Network (CNN)-based object detector such as YOLO locates and classifies potholes in real time [2], [9]. Many current platforms combine this detection capability with GPS geo-tagging, cloud storage, and a citizen- or municipality-facing dashboard, turning a simple detection algorithm into an end-to-end reporting and maintenance-management system [13], [14].

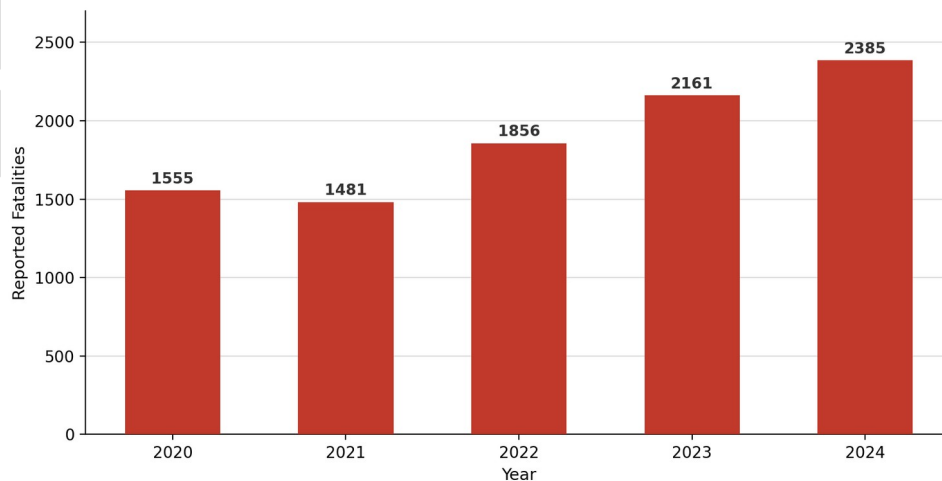
This paper does not propose or implement a new pothole-detection system. Instead, in line with the assigned project-documentation task, it studies and reviews several already-existing AI/ML-based pothole detection and reporting projects — spanning published research (YOLO-based detectors trained on Indian and multi-national road datasets), open-source citizen applications, and commercial smart-city platforms — and consolidates their underlying theory, architecture, working process, and reported performance into a single technical reference. The intent is for this review to serve as the theoretical and technical foundation on which a project such as "StreetGuard AI Pothole Reporter" can be designed.

The remainder of the paper is organised as follows: Section 2 discusses the need for such a project, supported by accident and cost statistics; Section 3 reviews specific existing systems studied for this report; Section 4 lists the technologies and techniques these systems commonly use; Section 5 explains the core scientific concepts (CNNs, YOLO, vibration sensing, and evaluation metrics); Section 6 describes a generalised system architecture and methodology; Section 7 walks through the working process step by step; Sections 8 and 9 cover datasets and reported results; and Sections 10 through 14 discuss advantages, benefits, applications, limitations, and future scope, before the paper concludes in Section 15.

2. Need of the Project

The scale of harm caused by potholes is large enough to justify automated, technology-driven solutions. According to figures placed before the Indian Parliament by the Ministry of Road Transport and Highways, pothole-related road accidents killed 9,438 people in India between 2020 and 2024 — a 53% rise over the five-year period — with the annual toll climbing from 1,555 deaths in 2020 to 2,385 in 2024. Uttar Pradesh alone accounted for more than half of these fatalities. Over the same period, 23,056 pothole-related accidents were reported nationally, resulting in 19,956 injuries, of which 9,670 were classified as grievous [29], [30], [31]. Figure 5 plots the year-wise fatality trend.

Fig. 5 Pothole-Related Road Accident Deaths in India, 2020-2024



Source: Ministry of Road Transport & Highways (MoRTH) data tabled in the Lok Sabha, reported Feb. 2026.

Fig. 5. Pothole-related road accident deaths in India, 2020–2024 (Source: MoRTH data tabled in the Lok Sabha, reported February 2026) [29]–[31].

The problem is not confined to India. In the United States, pothole-related vehicle damage cost motorists more than USD 26.5 billion in 2021 alone, and a 2022 AAA survey found that roughly one in ten drivers had to repair damage caused by a pothole; pavement distress of this kind also

contributes to an estimated 500 fatal accidents every year in the country. The American Society of Civil Engineers' 2025 infrastructure report card further noted that about 39% of public roads in the United States are in poor-to-mediocre condition [20].

Beyond the raw numbers, the way potholes are currently managed compounds the problem. Manual inspection is labour-intensive and cannot keep pace with a growing road network and vehicle fleet. Reactive, complaint-based systems suffer from under-reporting because many citizens do not bother to file a report, and even when complaints are filed, repair priority is often decided by how frequently a location is complained about rather than by an objective measure of severity, so that genuinely dangerous potholes in low-traffic or low-literacy areas may be neglected [13], [15]. Location data supplied through phone calls or informal descriptions is frequently imprecise, further delaying repair crews.

An AI/ML-based detection and reporting system addresses each of these gaps directly: it can survey roads continuously and automatically (rather than periodically and manually), it can assign a repeatable, data-driven severity score instead of relying on complaint volume, it can attach exact GPS coordinates to every detection, and it can make the resulting data available in real time to both citizens and road authorities through a shared dashboard. This combination of continuous monitoring, objective prioritisation, and transparent tracking is the central motivation for projects in this space, including "StreetGuard AI Pothole Reporter."

3. Literature Review: Existing Systems Studied

This section summarises five representative, already-existing AI/ML-based pothole detection and reporting efforts that were studied for this review. They were chosen to cover the main design philosophies found in the field: pure vision-based deep learning, sensor/vibration-based detection, and hybrid crowdsourced citizen-reporting platforms.

3.1 Vision-Based Pothole Detection and Reporting System for Smart Cities (YOLOv8-Seg + Predictive Maintenance)

Presented at a smart-cities conference in Pune, India, in December 2025, this system is the closest published analogue to "StreetGuard AI Pothole Reporter" and was used as the primary case study for this review. It is organised into four layers: a detection layer that runs a YOLOv8-Seg segmentation model (with ESRGAN-based image enhancement applied first to compensate for poor lighting or weather); a depth-and-severity analysis layer that uses monocular depth-estimation models (MiDaS and the Dense Prediction Transformer) together with an XGBoost regressor to combine pothole width, estimated depth, and local traffic density into a single

severity score; a geo-tagging and cloud-integration layer that stamps every detection with a Google Maps location and pushes the image and score to a Firebase/AWS backend; and a predictive-maintenance layer that applies ARIMA time-series forecasting to historical weather, traffic, and road-condition data to flag emerging "hot spots" before a full pothole forms [13].

3.2 Spothole — AI-Powered Citizen Pothole Reporting App

Spothole is an open-source citizen-facing Android/web application in which a user photographs a stretch of road; the image is passed to a Convolutional Neural Network trained for object detection/tracking, which geocodes and validates the presence of potholes in the scene. Verified reports are shown on a public map, and a dynamic ticket is generated for the nearest municipal authority, who can manage the report through their own jurisdiction-based dashboard. The project explicitly cites the scale of the problem — including widely reported figures of thousands of pothole-related deaths annually in India — as its motivating rationale [14].

3.3 RoadMetrics — AI-Based Road Health Survey Platform

RoadMetrics, a Bengaluru-based startup, mounts a smartphone on a vehicle's windshield and records video together with GPS coordinates and timestamps as the vehicle is driven normally. The video is uploaded to cloud servers where a trained computer-vision model identifies road defects; the company reports that its model can recognise ten categories of road distress ranging from minor surface cracking to full potholes. Unlike a purely academic prototype, RoadMetrics operates a business-to-business model, selling road-health survey reports to municipalities, highway authorities, and private fleet operators, and has been used to survey tens of thousands of kilometres of urban road in India [18].

3.4 Street Bump — Sensor-Based Crowdsourced Detection

One of the earliest large-scale deployments, the City of Boston's Street Bump app (2011) used an Android phone's built-in accelerometer and GPS while mounted in a moving vehicle: a sudden vertical jolt above a threshold was logged as a candidate pothole event, and events reported independently by multiple vehicles at the same location were treated as more reliable. City officials later acknowledged that the approach produced a significant number of false positives — the system sometimes could not distinguish a genuine pothole from a railway crossing — which motivated later work on smarter, learning-based filtering rather than a simple fixed threshold [17].

3.5 Academic YOLO-Based Detectors and the RDD Dataset Family

A large body of academic work has trained YOLO variants specifically for pothole detection. Dharneeshkar et al. built a 1,500-image dataset of Indian road conditions (captured in Coimbatore and Idukki under varied lighting, angle, and weather) and trained a YOLO model that achieved a mean Average Precision (mAP) of 72.12%, demonstrating the value of region-specific training data for Indian road conditions [2], [3]. At a larger scale, the Crowdsensing-based Road Damage Detection Challenge (CRDDC'2022) released the RDD2022 dataset — 47,420 images and over 55,000 annotated instances of road damage from six countries — as a public benchmark; the best-performing model in that challenge achieved an F1 score of 76.9% across all six countries [1]. These efforts illustrate how the field has progressively moved from small, single-region datasets toward large, multi-national, standardised benchmarks.

4. Technologies and Techniques Used

Although the systems reviewed in Section 3 differ in scope and deployment model, they draw on a broadly common technology stack. This section groups the techniques and tools reported across the literature into functional categories.

4.1 Programming Languages and ML/DL Frameworks

- Python — used almost universally for model training, data pre-processing, and experimentation.
- TensorFlow and Keras, and PyTorch — deep-learning frameworks used to build and train CNN and YOLO-based models.
- Ultralytics YOLOv5 / YOLOv8 libraries — high-level, ready-to-train implementations of the YOLO family used in several of the reviewed projects [6], [7], [8].
- Java / Kotlin (native Android) and Flutter / Dart (cross-platform) — used to build the citizen-facing mobile applications, e.g. Spothole and Civic Alerts [14], [15].

4.2 Computer Vision and Object Detection Techniques

- OpenCV — for frame extraction, image resizing, colour-space conversion, and classical image-processing pre-filters.
- Single-shot object detectors (YOLO family: v3, v4, v5, v8) — the dominant choice for real-time pothole detection because they trade a small amount of accuracy for very high inference speed [9], [10], [11].

- Two-stage detectors (Faster R-CNN, SSD) — used in some comparative studies as a slower but sometimes more accurate baseline against YOLO [10].
- Segmentation models (YOLOv8-Seg) — used where the exact area/outline of a pothole, not just a bounding box, is needed to estimate size and severity [7], [13].
- ESRRGAN (super-resolution GAN) — applied as a pre-processing step to sharpen low-light or low-resolution frames before detection [13].
- MiDaS / Dense Prediction Transformer — monocular depth-estimation networks used to infer pothole depth from a single 2-D image [13].

4.3 Sensor and Edge Hardware

- MEMS accelerometers and gyroscopes — either the ones built into an ordinary smartphone or a dedicated module (e.g. ADXL335) — used to sense the vertical jolt and tilt produced when a wheel crosses a pothole [17], [21], [23], [25].
- GPS / GNSS modules — for geo-tagging every detection with latitude and longitude [22], [25].
- Microcontrollers such as NodeMCU-ESP8266 — used in low-cost IoT prototypes to combine accelerometer, GPS, and Wi-Fi upload on a single board [25].
- Edge AI hardware (Raspberry Pi, NVIDIA Jetson Nano, dash-cam-class ADAS cameras) — used where inference needs to run on-vehicle rather than in the cloud, for lower latency and offline capability [20].

4.4 Backend, Cloud, and Mapping Technologies

- Cloud back-ends such as Firebase and AWS — for storing images, metadata, and severity scores, and for serving the mobile/web dashboard [13], [15].
- Relational and NoSQL databases (MySQL, PostgreSQL, Firestore) — for structured storage of tickets, statuses, and user accounts.
- REST APIs — for communication between the mobile app, the AI inference service, and the dashboard back-end.
- Google Maps API / GIS layers — for plotting detections on a map and computing proximity to known infrastructure [13], [16].
- ARIMA and gradient-boosted models (XGBoost) — applied to historical detection data for hotspot prediction and severity scoring rather than image analysis itself [13].

4.5 Frontend and User-Facing Technologies

- Responsive web dashboards (commonly built with React or similar frameworks) for municipal administrators to view, assign, and close tickets [15].
- Native or cross-platform mobile apps for citizens to capture and submit reports, track status, and receive notifications [16], [34].

5. Scientific Concepts Behind the System

This section explains, at a conceptual level, the core science that makes AI/ML-based pothole detection possible: how a CNN learns to "see" a pothole, how the YOLO family turns that ability into fast object detection, how vibration-based sensing offers a lightweight alternative, and how the resulting models are objectively evaluated.

5.1 Convolutional Neural Networks (CNNs)

A CNN is a deep-learning architecture designed specifically for grid-like data such as images. Instead of connecting every input pixel to every neuron (which would be computationally infeasible for a full-resolution photograph), a CNN slides small learnable filters, or kernels, across the image. Each filter produces a feature map that responds strongly wherever a particular local pattern — an edge, a corner, a patch of a certain texture or colour — is present. A Rectified Linear Unit (ReLU) activation is applied after each convolution to introduce non-linearity, and pooling layers (typically max-pooling) periodically down-sample the feature maps, which both reduces computation and makes the learned features more tolerant to small shifts in the object's position. Stacking several such convolution-activation-pooling blocks allows early layers to learn simple features like edges and textures, while deeper layers combine these into increasingly abstract shapes — in the case of a road surface, patterns such as the dark, irregular, shadow-edged region typical of a pothole. The final feature maps are flattened and passed through one or more fully-connected layers that combine all the learned features into a prediction, with the whole network trained end-to-end using labelled images and back-propagation. Figure 2 illustrates this pipeline.

Fig. 2 Convolutional Neural Network (CNN) Pipeline for Pothole Feature Learning

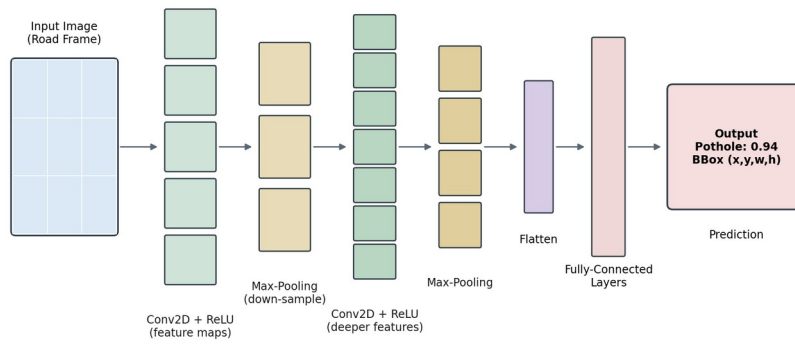


Fig. 2. Convolutional Neural Network (CNN) pipeline for pothole feature learning.

5.2 Object Detection with YOLO (You Only Look Once)

Image classification alone only answers "is there a pothole in this image?" — a usable pothole detector must also say where the pothole is. YOLO solves this localisation-plus-classification problem in a single forward pass through one network, which is why it is described as a single-shot detector and why it is fast enough for real-time video. YOLO divides the input image into an $S \times S$ grid; the grid cell in which the centre of a pothole falls is made "responsible" for detecting it. That cell predicts a fixed number of candidate bounding boxes, each described by its centre coordinates (x, y) , width, height, and a confidence score reflecting how likely it is that the box contains a real object and how accurate the box's shape is, together with class probabilities (in the simplest case, just "pothole" versus "background"). Because several overlapping boxes can end up describing the same pothole, a Non-Maximum Suppression (NMS) step discards all but the highest-confidence box in each cluster of overlapping predictions. Figure 3 illustrates the grid-and-bounding-box concept. Successive YOLO versions (v1 through the current v8/v11-class models) have refined this basic idea with deeper backbones, anchor boxes, batch normalisation, and — in the case of YOLOv8-Seg used by the Section 3.1 case study — an additional segmentation head that outputs a pixel-level mask rather than just a rectangular box, which is useful for estimating a pothole's exact area [26]–[28].

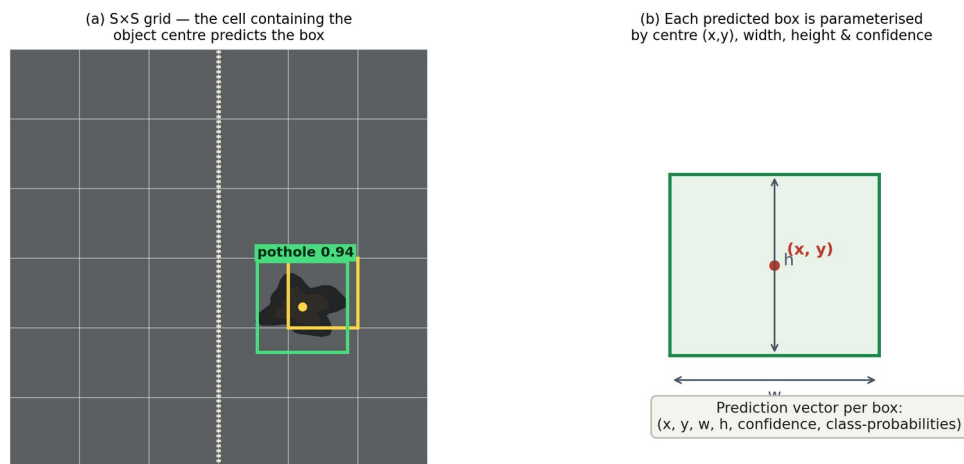
Fig. 3 Core Concept of Grid-Based Single-Shot Detection (YOLO)

Fig. 3. Core concept of grid-based single-shot detection (YOLO): (a) the responsible grid cell; (b) bounding-box parameters.

5.3 Vibration / Sensor-Based Detection

An alternative to analysing camera images is to sense the physical effect of a pothole on a moving vehicle. A three-axis accelerometer measures acceleration along the x , y , and z axes; when a wheel drops into and climbs out of a pothole, the vertical (z -axis) acceleration shows a brief, sharp spike, while a gyroscope simultaneously records the accompanying change in the vehicle's tilt or angular velocity. The simplest detection algorithms, such as the Z-THRESH and Z-DIFF methods used in early systems, simply flag an event whenever this signal crosses a fixed threshold; more robust implementations use a dynamic or statistically-derived threshold (for example, based on a Gaussian model of "normal" road vibration) to reduce false alarms caused by speed bumps, manhole covers, or rough-but-safe road texture [21], [24]. More recent work goes a step further by treating the vibration signal itself as an image (a pixel-based visual representation of the raw waveform) and classifying it with a CNN; one such study reports a classification accuracy of 93.24% using this approach, appreciably higher than a simple fixed-threshold method [21].

5.4 Evaluation Metrics

Because different studies use different models and datasets, a small set of standard metrics is used to compare them on a common footing.

- **Intersection over Union (IoU):** the ratio of the overlapping area between a predicted bounding box and the ground-truth box to the area of their union; a detection is usually only counted as correct if its IoU with the true pothole exceeds a threshold (commonly 0.5).

- **Precision:** the proportion of predicted potholes that are genuine potholes, i.e. $\text{True Positives} \div (\text{True Positives} + \text{False Positives})$ — a measure of how many of the alarms raised were real.
- **Recall:** the proportion of genuine potholes that the model successfully detects, i.e. $\text{True Positives} \div (\text{True Positives} + \text{False Negatives})$ — a measure of how few real potholes were missed.
- **Mean Average Precision (mAP):** the precision-recall curve is summarised as an Average Precision (AP) value for each object class, and mAP is the mean of these AP values, usually reported at a specific IoU threshold (e.g. mAP@0.5); it is the single most widely reported headline metric in the pothole-detection literature.
- **F1-Score:** the harmonic mean of precision and recall, useful when a single balanced number is needed — this is the metric used to rank submissions in the CRDDC'2022 road-damage challenge [1].
- **Frames Per Second (FPS):** a measure of inference speed rather than accuracy, indicating whether a model is fast enough for real-time deployment on live video.

6. System Architecture and Methodology

While every deployed system differs in implementation detail, the platforms reviewed in Section 3 share a common layered architecture, generalised in Figure 1.

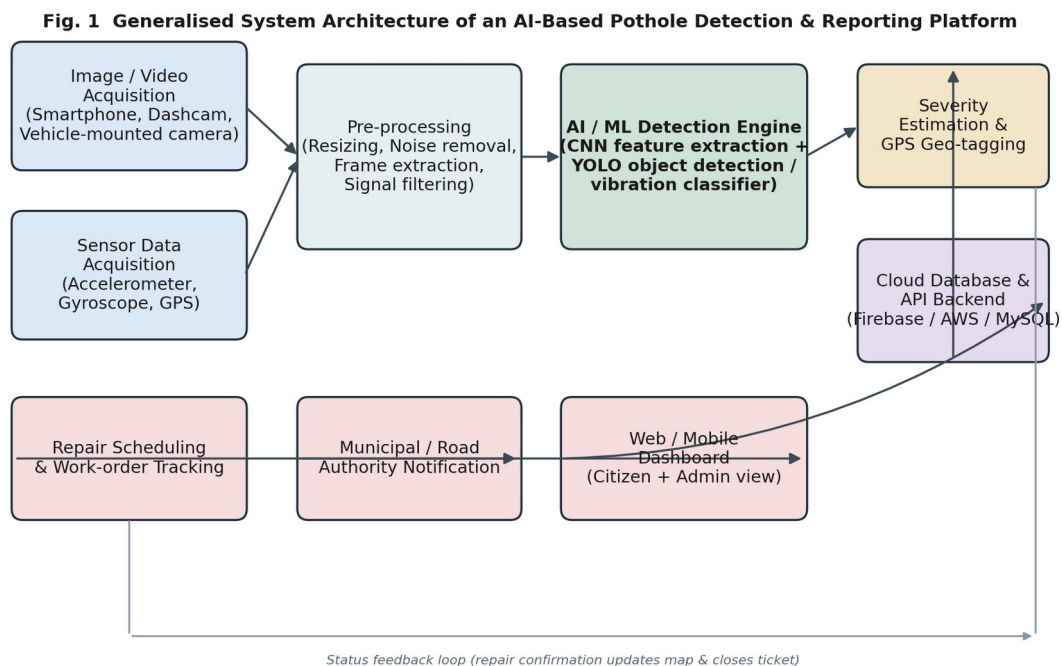


Fig. 1. Generalised system architecture of an AI-based pothole detection and reporting platform.

6.1 Architectural Layers

- **Data acquisition layer:** a smartphone or vehicle-mounted camera captures continuous road-surface video, while an accelerometer, gyroscope, and GPS module (where fitted) simultaneously log motion and location data.
- **Pre-processing layer:** video frames are resized, denoised, and colour-normalised, and low-light or low-resolution frames may be enhanced (e.g. with ESRGAN); vibration signals are filtered to remove sensor noise.
- **AI detection engine:** a CNN backbone extracts visual features, and a YOLO-style detection head (or, on the sensor side, a vibration classifier) outputs candidate pothole locations with confidence scores; Non-Maximum Suppression removes duplicate detections.
- **Severity estimation and geo-tagging layer:** bounding-box size, segmentation area, or estimated depth is converted into a severity score, and the detection is stamped with the synchronised GPS coordinate.
- **Cloud database and API back-end:** detection metadata (image, location, severity, timestamp) is uploaded to a cloud database (e.g. Firebase or AWS) over the device's network connection.
- **Presentation / dashboard layer:** a web and/or mobile dashboard visualises pothole locations on a map, lets citizens track the status of their reports, and gives municipal staff a work-order view.
- **Feedback loop:** once a repair crew closes a work order, the record is marked resolved and the public map layer is updated, closing the loop back to citizens.

6.2 Model Training Methodology

The AI detection engine itself is built through a fairly standard supervised-learning pipeline, described consistently across the reviewed literature:

- **Data collection** — road images or video frames are gathered from smartphones, dash-cams, or public datasets, ideally covering a range of lighting, weather, road-surface, and geographic conditions [2], [3].
- **Annotation** — each pothole in each image is manually or semi-automatically outlined with a bounding box or polygon, using tools such as Labellmg or Roboflow, and saved in a standard format (PASCAL VOC XML, YOLO .txt, or COCO JSON).
- **Train / validation / test split** — the annotated dataset is divided (a common ratio is roughly 70/20/10) so that model performance can be measured on data the model has not seen during training.

- Transfer learning — rather than training a detector from scratch, most projects start from weights pre-trained on a large general-purpose dataset (e.g. COCO) and fine-tune only on the pothole-specific data, which greatly reduces the amount of labelled data and training time required.
- Hyper-parameter tuning — learning rate, batch size, input resolution, and the number of training epochs are adjusted to balance accuracy against training time and the risk of over-fitting.
- Evaluation and iteration — the model is scored on the held-out test set using the metrics defined in Section 5.4, and the cycle (more data, different augmentation, or an alternative backbone) is repeated until performance is acceptable for deployment.

6.3 Database Configuration and Schema Design

Behind the dashboard layer, every system reviewed in Section 3 relies on a database to turn a one-off detection into a trackable record. Figure 6 presents a representative schema, consistent with the Firebase / AWS / MySQL-class back-ends referenced in Section 4.4, generalised from the architectures studied rather than tied to any single deployment.

Fig. 6 Representative Database Schema for an AI-Based Pothole Detection & Reporting Platform

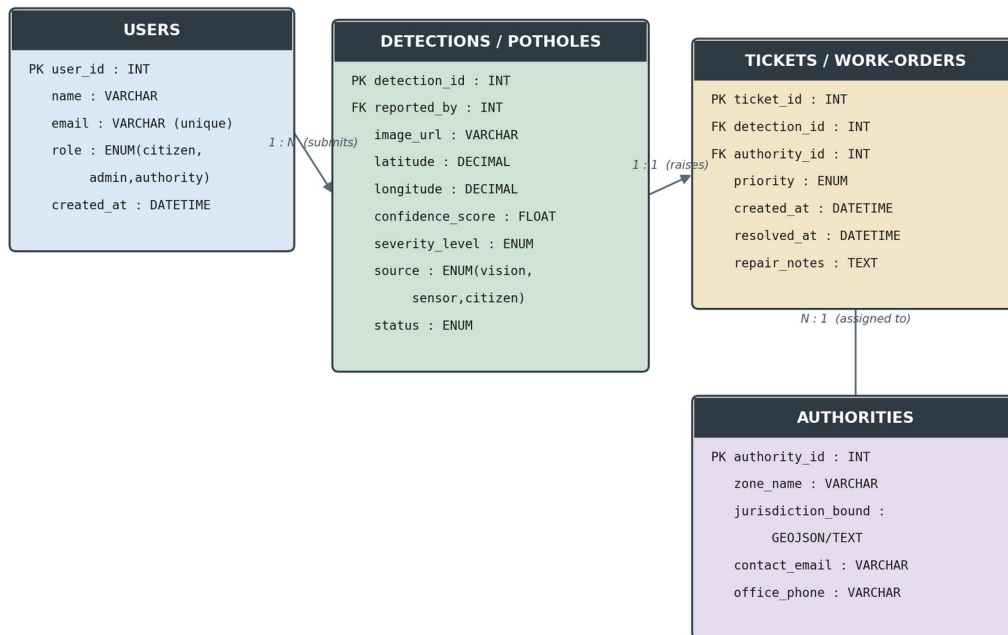


Fig. 6. Representative database schema for an AI-based pothole detection and reporting platform.

Four tables are sufficient to describe the core data model. The **USERS** table holds citizen and administrator accounts, distinguished by a role field. The **DETECTIONS / POTHOLES** table is the central record: each row stores the image reference, GPS latitude/longitude, the model's

confidence score, an assigned severity level, whether the detection originated from vision, sensor, or manual citizen upload, and a status flag. Each verified detection can raise one row in the TICKETS / WORK-ORDERS table, which links back to the originating detection and forward to the AUTHORITIES table representing the municipal zone or department responsible for the repair, along with priority, timestamps, and free-text repair notes. This structure keeps the raw AI output (a detection) cleanly separated from the human/administrative workflow (a ticket), which is what allows the same detection data to support both a public map view and an internal work-order queue.

6.4 Sample Input-Output Execution Walkthrough

To make the abstract pipeline in Figure 1 concrete, Figure 7 walks through one illustrative detection end to end, synthesising the input, processing, and output stages as they are described across the reviewed systems.

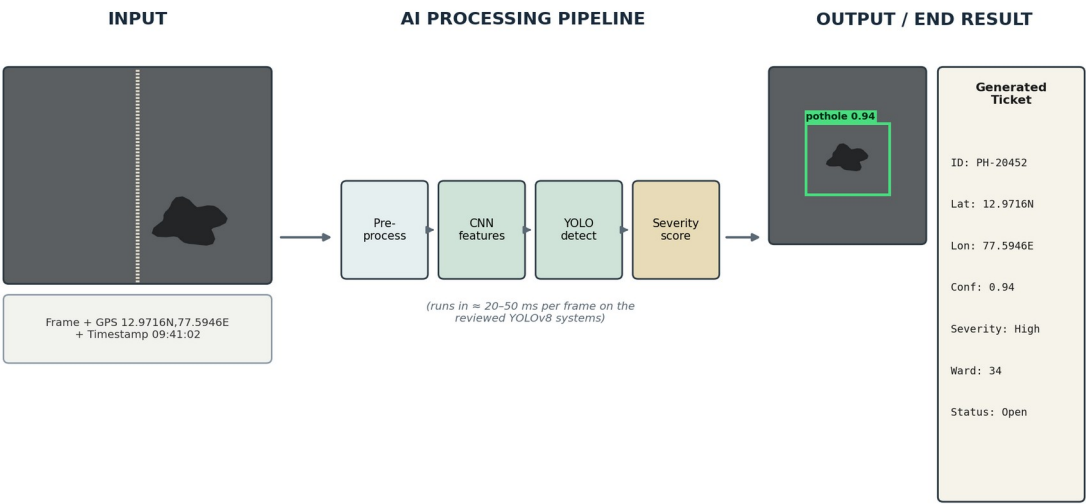


Fig. 7 Sample Input-Processing-Output Execution Walkthrough (illustrative example based on the systems reviewed in Section 3)

Fig. 7. Sample input-processing-output execution walkthrough (illustrative example).

The input to the system is simply a road-surface video frame captured by a smartphone or dash-cam, paired with the GPS coordinate and timestamp recorded at the same instant. This pair is what allows every later step to be geo-referenced. The frame then passes through the processing pipeline described in Sections 5 and 6.1: pre-processing normalises the image, the CNN backbone extracts features, the YOLO detection head proposes a bounding box and confidence score, and the severity-estimation logic converts box size (and, where available, estimated depth) into a severity label. On the reviewed YOLOv8-class systems this entire chain completes in roughly 20–50 milliseconds per frame [8], fast enough to run continuously on live video.

The end result delivered back to the user is two-fold: an annotated image with the pothole outlined and labelled with its confidence score (useful for human verification), and a structured ticket record — a unique detection ID, the GPS coordinate, confidence, severity, the municipal ward it has been routed to, and a status field that starts at "Open" and is updated as the record moves through the workflow in Figure 6. This ticket is what ultimately appears as a marker on the public dashboard described in Section 6.1 and is what a maintenance crew acts on in the final "Repair & Closure" step of Figure 4.

7. Working Process

Figure 4 traces the end-to-end working process of a typical AI-based pothole detection and reporting platform, from the moment a vehicle passes over a road to the moment a repair is confirmed.

Fig. 4 Step-by-Step Working Process of the AI-Based Pothole Detection & Reporting Pipeline



Fig. 4. Step-by-step working process of the AI-based pothole detection and reporting pipeline.

In practice, most of this pipeline runs automatically and continuously while the host vehicle or app is active. Steps 1–4 (capture through post-processing) typically execute on-device or at the edge, within tens of milliseconds per frame, so that detection keeps pace with live video — the YOLOv8-based detector reviewed in Section 3.5, for instance, is reported to process a single image in about 22 milliseconds [8]. Steps 5–7 (severity scoring, upload, and dashboard notification) involve network communication and can tolerate slightly higher latency, since a repair decision does not need to be made in real time. The final step, repair and closure, is the one manual, human-driven stage in the pipeline: a municipal crew is dispatched based on the severity ranking shown on the dashboard, and once the work is physically completed, the ticket is marked resolved, which removes or updates the corresponding marker on the public map. This closes the loop between automated detection and verified, on-the-ground repair, which is the essential feature distinguishing a full detection-and-reporting platform from a bare detection model.

8. Dataset Description

The performance of any deep-learning-based pothole detector depends heavily on the size, diversity, and quality of the dataset it is trained on. Table 1 summarises the principal datasets referenced in the reviewed literature.

Dataset	Size	Coverage	Damage Classes
RDD2022 [1]	47,420 images; 55,000+ annotated instances	Japan, India, Czech Republic, Norway, USA, China	Longitudinal crack, transverse crack, alligator crack, pothole
RDD2020 [1]	26,336 images; 31,000+ instances	India, Japan, Czech Republic	Same four classes as RDD2022 (predecessor dataset)
Custom Indian Road Dataset [2],[3]	1,500 images	Coimbatore & Idukki, India	Pothole (single class), varied lighting/weather/angle
Roboflow Public Pothole Dataset [6]	Open, community-sourced	Global (crowd-contributed)	Pothole
Kaggle Pothole Image Dataset [10]	665 images	Global	Pothole

Table 1. Representative datasets used for training AI/ML-based pothole detectors.

RDD2022 is particularly significant because it was released as part of the Crowdsensing-based Road Damage Detection Challenge (CRDDC'2022), an IEEE Big Data Cup that invited researchers worldwide to benchmark detection algorithms on a single, standardised, multi-national dataset — moving the field away from small, single-city datasets that generalise poorly to new geographies,

lighting conditions, and pavement types [1]. Smaller, region-specific datasets such as the 1,500-image Indian road set remain valuable, however, because road colour, texture, and typical pothole shape can differ noticeably by country, and a model trained only on, for example, Czech or Japanese roads may transfer poorly to Indian conditions without additional fine-tuning [2], [3].

9. Results and Performance Comparison

Table 2 collects the headline performance figures reported by the studies reviewed in Section 3, to give a sense of the accuracy and speed that current AI/ML-based pothole detectors achieve in practice.

Model / System	Dataset	Precision / Recall	mAP / F1 / Accuracy	Notes
YOLO (Dharneeshkar et al.) [2],[3]	Custom Indian roads, 1,500 imgs	—	mAP 72.12%	First large Indian-road-specific YOLO dataset
YOLOv4 (CSP-Darknet-53) [10]	Dash-cam images	P 85% / R 81%	mAP 85.39%	Benchmarked against SSD & YOLOv3
YOLO + haptic feedback app [5]	Custom pothole images	—	Accuracy 82.7%	30 FPS; aids visually-impaired navigation
YOLOv8 urban detector [8]	67 urban test images	High P & R (qualitative)	—	22.1 ms / image (~45 FPS)
CRDDC'2022 winning model [1]	RDD2022 (6 countries)	—	F1 76.9%	Best global entrant across all countries
YOLOv8n / v11n / v12n + GNSS/ROS [20]	Real-world uncontrolled routes	P 84% / R 84%	—	ADAS-integrated; varied weather & light
CNN on vibration-to-image [21]	Smartphone accelerometer + gyroscope	—	Accuracy 93.24% (loss 0.29)	Sensor-based, not vision-based

Table 2. Reported performance of representative pothole detection studies.

A few trends emerge from this comparison. First, vision-based deep-learning detectors generally report mAP or F1 figures in the 70–86% range on real-world road imagery, with accuracy strongly dependent on how closely the training data matches the deployment environment — the Indian-road-specific model [2] and the CRDDC'2022 multi-country winner [1] illustrate the gap between a narrow, single-region dataset and a broad, standardised one. Second, YOLO-family detectors consistently achieve real-time or near-real-time speeds (tens of milliseconds per frame), which is what makes them suitable for live-video deployment on a moving vehicle rather than offline batch analysis [8]. Third, the sensor/vibration-based CNN approach reports a notably high raw accuracy

figure (93.24%) [21], but this number is not directly comparable to the vision-based mAP/F1 figures because it is measured on a different task (binary event classification of a vibration segment, rather than localisation of a pothole within an image) and typically on a smaller, more controlled test set. In general, the literature suggests that vision-based methods offer richer information (location within the lane, approximate size, and even depth) at a higher computational cost, while vibration-based methods are cheaper and lower-power but historically more prone to false positives from other road features such as speed bumps or manhole covers [17], [24].

10. Advantages

Reviewing the systems in Section 3 against the manual/reactive baseline described in Section 2 highlights a consistent set of advantages offered by AI/ML-based pothole detection and reporting:

- Continuous, automated monitoring that does not depend on a human inspector physically visiting every stretch of road.
- Objective, repeatable severity scoring based on measured size, depth, or vibration intensity, rather than subjective judgement or complaint volume.
- Real-time or near-real-time detection speed (tens of milliseconds per frame for YOLO-based models), enabling live-video deployment on moving vehicles [8].
- Precise GPS geo-tagging of every detection, removing the ambiguity of phone-in or written complaints.
- Scalability — the same trained model can be deployed across an entire city or highway network at a marginal cost far lower than proportionally increasing human inspection staff.
- Transparency and accountability, since a shared dashboard lets citizens track the status of a reported or detected pothole through to repair [15], [16].
- Data-driven maintenance planning, allowing authorities to identify recurring hotspots and even predict where new damage is likely to appear, as in the ARIMA-based predictive-maintenance layer described in Section 3.1 [13].
- Extensibility to related road-safety uses, since the same camera pipeline can often be adapted to detect other pavement distress types (cracks, potholes of varying severity, debris) with additional training data [18].

11. Benefits

11.1 Benefits to Citizens and Road Users

- Reduced risk of accidents, injury, and vehicle damage from unexpected potholes.
- A simple, low-effort way to report hazards (a photo and automatic GPS tag) instead of navigating a complaint office or call centre [16].
- Visibility into whether a reported hazard is being acted upon, which builds public trust in civic institutions [15].

11.2 Benefits to Government and Municipal Bodies

- Objective data to prioritise limited repair budgets toward the most severe and highest-traffic locations first.
- Lower long-term cost of road maintenance, since early repair of a small pothole is far cheaper than reconstructing a badly deteriorated section later.
- A verifiable audit trail of detection, assignment, and repair, which is useful for accountability and, in some jurisdictions, for defending against compensation claims [30].

11.3 Economic and Societal Benefits

- Reduced vehicle-repair expenditure at a national scale — pothole damage alone was estimated at over USD 26.5 billion in the United States in 2021 [20].
- Fewer traffic disruptions and delays caused by vehicles slowing or swerving to avoid road damage.
- Support for broader smart-city initiatives, since the same sensing and dashboard infrastructure can often be extended to other civic issues such as garbage accumulation or streetlight failure [15], [23].

11.4 Safety Benefits

- Direct reduction in the pothole-related fatalities and injuries documented in Section 2, if detections are consistently converted into timely repairs.
- Assistive applications for vulnerable road users, such as the YOLO-based system that provides auditory/haptic pothole alerts to visually impaired pedestrians [5].

12. Applications

Beyond simple citizen reporting, the core technology reviewed in this paper has been applied — or is directly applicable — across several domains:

- Municipal and highway road-maintenance management, where detections feed directly into work-order and budgeting systems [13], [18].
- Smart-city infrastructure monitoring, integrating road-condition data with GIS platforms alongside other civic sensors [16].
- Advanced Driver-Assistance Systems (ADAS) and autonomous vehicles, where real-time pothole detection contributes to path-planning and ride-comfort decisions [10], [20].
- Fleet management and logistics, where companies use dash-cam-based detection to protect vehicles and plan routes that avoid known road damage [19], [27].
- Insurance and vehicle-damage assessment, using historical road-condition data to help verify damage claims.
- Assistive technology for visually impaired pedestrians, using real-time detection to provide auditory or haptic warnings [5].
- Highway and pavement asset-management systems, where periodic AI-based road surveys replace or supplement manual condition surveys used for long-term asset planning [32].

13. Limitations and Challenges

Despite their advantages, the systems reviewed also reveal a consistent set of limitations that any similar project, including StreetGuard AI Pothole Reporter, would need to plan for:

- Dataset bias and limited geographic generalisation — a model trained on one country's roads (surface colour, pothole shape, signage) can under-perform when deployed elsewhere without additional fine-tuning [2], [3].
- Sensitivity to lighting and weather — rain, glare, fog, and low light all degrade detection accuracy unless specifically compensated for, e.g. through image-enhancement pre-processing [12], [13].
- False positives from visually or mechanically similar features — shadows, manhole covers, tar patches, and railway crossings can be mistaken for potholes by both vision- and vibration-based systems [17].

- Compute constraints on edge devices — running a full-accuracy deep model in real time on a low-cost smartphone or embedded board may require model compression or a smaller/faster network variant.
- Occlusion and partial visibility — potholes partly hidden by water, debris, or other vehicles can be missed or mis-sized.
- Privacy considerations around continuous GPS tracking and camera footage of public roads, which need to be addressed through data-handling policy.
- Connectivity gaps in rural or low-network-coverage areas, which can delay upload of detections to a cloud back-end.
- Lack of a universally standardised severity scale, which makes it hard to compare or merge data from different vendors or research groups.
- Integration friction with legacy municipal ticketing and works-management systems that were not designed to ingest automated, high-volume data feeds.

14. Future Scope

The literature reviewed points to several directions in which AI/ML-based pothole detection and reporting is likely to continue evolving:

- Multi-sensor fusion, combining vision, vibration, and depth/LiDAR data to reduce false positives and improve severity estimation beyond what any single sensing mode can achieve alone [24].
- Wider deployment of edge AI on 5G-connected dash-cams and fleet vehicles, enabling on-vehicle inference with cloud upload only for confirmed, high-confidence detections [19], [20].
- Drone- and satellite-based aerial road surveys for rapid, large-scale condition assessment of highway networks that are impractical to survey continuously by ground vehicle.
- Predictive, rather than purely reactive, maintenance — extending time-series approaches like the ARIMA-based hotspot forecasting seen in Section 3.1 so that road segments likely to develop potholes are resurfaced pre-emptively [13].
- Digital-twin representations of road networks that continuously ingest detection data to simulate wear patterns and optimise maintenance scheduling.
- Tighter integration with autonomous-vehicle path planning, so that detected road damage directly informs real-time route and trajectory decisions [10].

- Use of synthetic and generative data augmentation to enlarge and diversify training datasets for regions or conditions that are currently under-represented, addressing the geographic-bias limitation discussed in Section 13.
- Transparent, verifiable public tracking of repair status — potentially using tamper-evident logging — to further strengthen citizen trust in municipal response [15].

15. Conclusion

This review has examined the theory, architecture, working process, and reported performance of several already-existing AI/ML-based pothole detection and reporting systems, rather than building a new one from scratch, in line with the assigned project-documentation objective. Across the systems studied — from early accelerometer-based crowdsourcing such as Street Bump, through CNN/YOLO-based vision detectors trained on datasets ranging from a 1,500-image Indian road set to the 47,420-image multi-national RDD2022 benchmark, to full citizen-reporting platforms such as Spothole and the YOLOv8-Seg-based smart-city system studied as the primary case study — a common pattern emerges: automated sensing (vision and/or vibration), followed by AI-based detection and severity scoring, GPS geo-tagging, cloud storage, and a shared dashboard that closes the loop between detection and verified repair.

Given the scale of the problem documented in Section 2 — thousands of preventable deaths and billions of dollars in vehicle damage annually — the case for automating pothole detection and reporting is strong, and the reviewed literature shows that current deep-learning and sensor-based methods are already accurate and fast enough for practical deployment, even as challenges around geographic generalisation, false positives, and system integration remain active areas of work. This body of existing theory and practice provides a solid technical foundation on which the "StreetGuard AI Pothole Reporter" project can be designed, positioned, and evaluated.

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